

Stable Marriage, Optimal Transport, and Birkhoff Geometry

Ordinal stability, cardinal efficiency, and transferable utility

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Abstract

Stable marriage and discrete optimal transport both pair two finite populations, but they answer different questions. Gale–Shapley stability is ordinal and local: no unmatched pair should prefer one another to their assigned partners. Kantorovich transport is cardinal and global: it minimizes an aggregate cost over the Birkhoff polytope. This note separates these notions before drawing their precise connections. The Birkhoff–von Neumann theorem identifies the Birkhoff polytope with the convex hull of all perfect matchings, while the Vande Vate–Rothblum theorem identifies a smaller polytope, obtained by adding linear no-blocking inequalities, with the convex hull of stable matchings. Ordinary stability is not equivalent to minimum total cost, even when both sides rank partners using the same cost matrix. The exact optimal-transport bridge instead appears in the Shapley–Shubik assignment game: with transferable utility, pairwise stability is Kantorovich dual feasibility, matched pairs satisfy complementary slackness, and Birkhoff integrality yields an integral stable assignment. We also explain what Gale–Shapley optimizes, compare its algorithmic invariants with Hungarian primal–dual methods, and discuss unequal populations, fractional masses, and continuous matching.

1 Three questions about the same pairing matrix

Let $X = \{1, \dots, n\}$ and $Y = \{1, \dots, n\}$ be two populations. A perfect matching is a permutation $\sigma \in \mathfrak{S}_n$: agent $i \in X$ is paired with $\sigma(i) \in Y$. Its permutation matrix is

$$(P_\sigma)_{ij} = \begin{cases} 1, & j = \sigma(i), \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

This common representation hides three different mathematical models.

- 1. Optimal assignment.** A cardinal cost C_{ij} measures the loss incurred by matching i to j . One minimizes the sum of costs.
- 2. Stable marriage.** Each agent only ranks the agents on the opposite side. One asks that no unmatched pair mutually prefer one another to their assigned partners.
- 3. Assignment game.** A match produces a cardinal surplus that the two partners can divide through a transfer. One asks that no pair can deviate and split its surplus so as to improve both partners' payoffs.

The first model is a social planner's optimization problem, the second is a no-blocking equilibrium without transfers, and the third is a market equilibrium with transferable utility. Confusing them leads to false claims. In particular, a stable marriage need not minimize any natural aggregate cost on all matchings. The assignment game, not ordinary stable marriage, is the setting in which Kantorovich duality becomes an exact stability theorem.

2 Optimal assignment and the Birkhoff polytope

We first recall the convex geometry underlying discrete transport. The Birkhoff polytope is

$$\mathcal{B}_n = \left\{ P \in \mathbb{R}_+^{n \times n} : P\mathbf{1} = \mathbf{1}, P^\top \mathbf{1} = \mathbf{1} \right\}. \quad (2)$$

Its elements are bistochastic matrices. If both empirical measures assign mass $1/n$ to every agent, the corresponding Kantorovich couplings are P/n with $P \in \mathcal{B}_n$. We retain the unnormalized convention (2) because its integral points are exactly the permutation matrices.

Definition 2.1 (Optimal assignment and its Kantorovich relaxation). For a cost matrix $C = (C_{ij}) \in \mathbb{R}^{n \times n}$, define

$$\text{Assign}(C) = \min_{\sigma \in \mathfrak{S}_n} \sum_{i=1}^n C_{i,\sigma(i)}, \quad (3)$$

$$\text{Kant}(C) = \min_{P \in \mathcal{B}_n} \langle C, P \rangle = \min_{P \in \mathcal{B}_n} \sum_{i,j} C_{ij} P_{ij}. \quad (4)$$

Theorem 2.2 (Birkhoff–von Neumann). *The extreme points of $\mathcal{B}_n$ are the permutation matrices. Equivalently, every $P \in \mathcal{B}_n$ admits a decomposition*

$$P = \sum_{k=1}^N \lambda_k P_{\sigma_k}, \quad \lambda_k \geq 0, \quad \sum_{k=1}^N \lambda_k = 1. \quad (5)$$

Consequently $\text{Kant}(C) = \text{Assign}(C)$, and the Kantorovich problem has an integral minimizer.

Proof. The support graph of a bistochastic matrix has no degree-one vertex unless its incident entry is one. If a nonintegral bistochastic matrix has no such forced edge, its bipartite support contains an even cycle. Alternately adding and subtracting a sufficiently small amount along that cycle writes the matrix as the midpoint of two distinct bistochastic matrices, so it is not extreme. Conversely, a permutation matrix cannot be written as a nontrivial convex combination of bistochastic matrices because each of its zero entries forces the corresponding entries of both matrices to vanish. This proves the extreme-point statement and hence (5). A linear functional reaches its minimum at an extreme point, proving the last claim. See [Birkhoff \(1946\)](#); [von Neumann \(1953\)](#). $\square$

The linear-programming dual of (4) is

$$\max_{f,g \in \mathbb{R}^n} \left\{ \sum_i f_i + \sum_j g_j : f_i + g_j \leq C_{ij} \text{ for all } i, j \right\}. \quad (6)$$

At optimality, complementary slackness gives

$$P_{ij} > 0 \implies f_i + g_j = C_{ij}. \quad (7)$$

The Hungarian algorithm is a primal–dual method built around precisely these feasible potentials and equality edges ([Kuhn, 1955](#)).

3 Stable marriage and deferred acceptance

Stable marriage replaces cardinal costs by ordinal preferences. For every $i \in X$, let $\succ_i^X$ be a strict total order on Y ; $j \succ_i^X j'$ means that i prefers j to j' . For every $j \in Y$, let $\succ_j^Y$ be a strict total order on X .

Definition 3.1 (Blocking pair and stable matching). For a matching $\sigma \in \mathfrak{S}_n$, an unmatched pair (i, j) , with $j \neq \sigma(i)$, is a *blocking pair* when

$$j \succ_i^X \sigma(i) \quad \text{and} \quad i \succ_j^Y \sigma^{-1}(j). \quad (8)$$

The matching is *stable* if it has no blocking pair.

Stability is local and ordinal. It does not sum gains across agents, and a large improvement for one pair cannot compensate for a loss to another pair. Gale and Shapley's deferred-acceptance algorithm proves that a stable matching always exists under strict complete preferences (Gale and Shapley, 1962).

The X -proposing deferred-acceptance algorithm. Initially all agents are unmatched. While an unmatched $i \in X$ has not proposed to everyone, i proposes to the most-preferred $j \in Y$ who has not yet rejected him. Each j tentatively holds her most-preferred proposal among her current tentative partner and all new proposals, and rejects the others. The tentative matches become final when no further proposal is possible.

Theorem 3.2 (Gale–Shapley). *With strict complete preferences, deferred acceptance terminates after at most n^2 proposals and returns a perfect stable matching σ_X . Moreover, it is X -optimal among stable matchings: for every stable σ and every $i \in X$,*

$$\sigma_X(i) \succeq_i^X \sigma(i). \quad (9)$$

If the roles are reversed, the resulting matching is Y -optimal.

Proof. No ordered pair (i, j) receives more than one proposal, so there are at most n^2 proposals. Once a receiver holds a proposal, she always holds one and her tentative partner can only improve. At termination every proposer is matched. Indeed, if some proposer had exhausted all receivers, every receiver would have received and hence would still hold one proposal, accounting for all n proposers.

To prove stability, suppose that (i, j) blocks the output. Since i prefers j to his final partner, he proposed to j earlier. Receiver j rejected him while holding someone she preferred, and her tentative partner only improved afterward. She therefore prefers her final partner to i , contradicting (8).

For proposer optimality, suppose for contradiction that some proposer is rejected by a receiver with whom he is paired in some stable matching. Consider the first such rejection: j rejects i while holding i' , whom she prefers to i . In a stable matching σ with $\sigma(i) = j$, write $j' = \sigma(i')$. Proposer i' must prefer j to j' . Otherwise he would already have proposed to, and been rejected by, his stable partner j' , contradicting the choice of the first rejection. Thus i' prefers j to j' , while j prefers i' to i , so (i', j) blocks σ . This contradiction shows that no proposer is ever rejected by a possible stable partner. His final partner is therefore his best stable partner, proving (9). Reversing the sides gives the last claim. $\square$

Stable matchings form a distributive lattice under the order induced by the X -agents' preferences; the two deferred-acceptance outcomes are its opposite extrema. This finer structure is developed by Gusfield and Irving (1989) and Roth and Sotomayor (1990).

4 A stable analogue of the Birkhoff polytope

The parallel with Kantorovich transport becomes geometric once stability is written as linear inequalities. For a pair (i, j) , set

$$B_i(j) = \{j' \in Y : j' \succ_i^X j\}, \quad B_j(i) = \{i' \in X : i' \succ_j^Y i\}. \quad (10)$$

Define the fractional stable-marriage polytope

$$\mathcal{S}(\succ^X, \succ^Y) = \left\{ P \in \mathcal{B}_n : P_{ij} + \sum_{j' \in B_i(j)} P_{ij'} + \sum_{i' \in B_j(i)} P_{i'j} \geq 1 \quad \forall (i, j) \right\}. \quad (11)$$

Proposition 4.1 (Integral no-blocking inequalities). *For $\sigma \in \mathfrak{S}_n$, its permutation matrix P_σ belongs to $\mathcal{S}(\succ^X, \succ^Y)$ if and only if σ is stable.*

Proof. For an integral matching, the left-hand side of the inequality indexed by (i, j) is positive precisely when at least one of the following holds: i is matched to j ; i is matched to someone preferred to j ; or j is matched to someone preferred to i . The inequality therefore fails exactly when i and j are unmatched and both prefer one another to their current partners, which is exactly the blocking condition. $\square$

The difficult statement is that the inequalities introduce no fractional extreme points. This is the stable-marriage counterpart of the Birkhoff–von Neumann theorem.

Theorem 4.2 (Vande Vate–Rothblum stable-polytope theorem). *Under strict complete preferences,*

$$\mathcal{S}(\succ^X, \succ^Y) = \text{conv} \{P_\sigma : \sigma \text{ is stable}\}. \quad (12)$$

In particular, every extreme point of (11) is the permutation matrix of a stable matching.

Proof structure. Proposition 4.1 gives the inclusion from right to left. The reverse inclusion is the nontrivial polyhedral theorem of Vande Vate (1989), simplified and extended by Rothblum (1992). Rothblum proves that every linear objective over the system (11) has an integral optimum. One can formulate the key step through the rotation poset of stable matchings: eliminating a closed set of rotations moves from the X -optimal matching to another stable matching, and the resulting order-polytope representation is integral. Equivalently, a linear image of that integral order polytope gives (11). Later, Király and Pap (2008) proved that Rothblum’s inequality system is totally dual integral.

For completeness, once this integrality result is granted, the convex-hull claim follows immediately. If (11) had a fractional extreme point P , choose a linear functional that uniquely exposes it. The integrality theorem would supply an integral optimizer for the same functional, contradicting uniqueness. Hence every extreme point is integral, and Proposition 4.1 identifies it with a stable matching. A bounded polytope is the convex hull of its extreme points. $\square$

Remark 4.3 (Two different integrality mechanisms). Theorem 2.2 says that every bistochastic matrix is a mixture of arbitrary perfect matchings. Theorem 4.2 says that every bistochastic matrix satisfying the fractional no-blocking inequalities is a mixture of stable matchings. An arbitrary Birkhoff decomposition of a point in $\mathcal{S}$ need not consist of stable permutations; the theorem guarantees that at least one stable decomposition exists.

Corollary 4.4 (Minimum-cost stable marriage). *For every weight matrix C , the linear program*

$$\min_{P \in \mathcal{S}(\succ^X, \succ^Y)} \langle C, P \rangle \quad (13)$$

has an integral optimum and therefore computes a minimum-cost stable matching.

This looks like Kantorovich transport, but the feasible polytope now depends on the two preference profiles. The cost only selects among stable matchings; it does not create stability.

What deferred acceptance optimizes. Let $r_{ij}^X \in \{1, \dots, n\}$ be the rank of j in proposer i 's preference list, with rank one best. The pointwise optimality (9) is stronger than minimizing one aggregate score.

Corollary 4.5 (A linear-programming characterization of Gale–Shapley). *For any weights $w_i > 0$, the X -proposing Gale–Shapley output solves*

$$\min_{P \in \mathcal{S}(\succ^X, \succ^Y)} \sum_{i,j} w_i r_{ij}^X P_{ij}. \quad (14)$$

It simultaneously minimizes every proposer's rank among stable matchings.

Proof. By Theorem 4.2, every feasible P is a convex combination of stable permutation matrices. For each such matching, (9) gives $r_{i, \sigma_X(i)}^X \leq r_{i, \sigma(i)}^X$ for every i . Multiplication by $w_i > 0$, summation, and convex combination prove the claim. $\square$

This does not imply that the X -optimal matching minimizes a two-sided rank sum or a cardinal transport cost over all matchings.

5 Why ordinary stability is not optimal transport

It is tempting to generate both preference profiles from a common cost matrix: each agent prefers the partner with smaller cost. Even then, no-blocking stability and minimum total cost generally disagree.

Example 5.1 (A stable matching can be globally expensive). Consider

$$C = \begin{pmatrix} 0 & 1 \\ 50 & 100 \end{pmatrix}, \quad (15)$$

and let every row and every column rank partners by increasing C_{ij} . The diagonal matching has total cost 100 and is stable. The pair $(1, 2)$ does not block because agent $1 \in X$ prefers $1 \in Y$, while $(2, 1)$ does not block because agent $1 \in Y$ prefers $1 \in X$. The off-diagonal matching has the smaller total cost $1 + 50 = 51$, so it is the unique optimal assignment, but the pair $(1, 1)$ blocks it. Thus the unique stable matching differs from the unique Kantorovich optimum.

There is also a geometric obstruction to encoding all stable matchings as the minimizers of a single unconstrained transport cost. The minimizers of a linear functional over $\mathcal{B}_n$ form a face of $\mathcal{B}_n$. The convex hull of the stable matchings is generally a polytope cut out by additional inequalities, not a face of $\mathcal{B}_n$. Hence there need not exist a matrix C whose entire set of optimal assignments is exactly the set of stable marriages.

The two notions answer different welfare questions:

- optimal transport permits losses and gains to compensate across unrelated agents through the sum $\sum_i C_{i,\sigma(i)}$;
- stable marriage forbids a deviation only when both members of one pair improve, and it allows no monetary compensation between them.

6 The exact bridge: transferable utility and the assignment game

The connection with Kantorovich duality becomes exact once utility is transferable. Let S_{ij} be the total surplus created by matching i and j . For simplicity assume that all agents are matched and that outside options have been normalized to zero. A matched pair may divide S_{ij} arbitrarily: u_i is the payoff of $i \in X$, and v_j that of $j \in Y$.

Definition 6.1 (Core-stable assignment-game outcome). An outcome (σ, u, v) is core stable when

$$u_i + v_{\sigma(i)} = S_{i,\sigma(i)} \quad \text{for every } i, \quad (16)$$

$$u_i + v_j \geq S_{ij} \quad \text{for every } (i, j), \quad (17)$$

with $u_i, v_j \geq 0$ when individual rationality relative to the outside option is required.

Indeed, pair (i, j) can make both members strictly better through a transfer if and only if its available surplus satisfies $S_{ij} > u_i + v_j$. Thus (17) is exactly the no-blocking condition with transfers.

Consider the primal and dual linear programs

$$\max_{P \in \mathcal{B}_n} \langle S, P \rangle, \quad (18)$$

$$\min_{u, v \in \mathbb{R}^n} \left\{ \sum_i u_i + \sum_j v_j : u_i + v_j \geq S_{ij} \quad \forall (i, j) \right\}. \quad (19)$$

Theorem 6.2 (Shapley–Shubik: core stability is Kantorovich duality). *An outcome (σ, u, v) is core stable if and only if P_σ solves (18), (u, v) solves (19), and the two solutions satisfy complementary slackness. Consequently the assignment game has a core-stable outcome.*

Proof. Suppose first that the outcome is core stable. The inequalities (17) make (u, v) dual feasible. Summing (16) over matched pairs yields

$$\langle S, P_\sigma \rangle = \sum_i S_{i,\sigma(i)} = \sum_i u_i + \sum_j v_j. \quad (20)$$

Weak duality says that every primal value is at most every dual value, so equality proves that both candidates are optimal. Equality on matched pairs is complementary slackness.

Conversely, let P_σ and (u, v) be optimal primal and dual solutions. Strong linear-programming duality equates their objectives. Complementary slackness gives $u_i + v_{\sigma(i)} = S_{i,\sigma(i)}$, while dual feasibility gives $u_i + v_j \geq S_{ij}$ for every pair. These are precisely (16)–(17).

Finally, linear-programming duality supplies a dual optimizer, and Theorem 2.2 supplies an integral primal optimizer. Together they form a core-stable outcome. This is the finite assignment-game theorem of [Shapley and Shubik \(1971\)](#). $\square$

To recover the cost convention, set $S_{ij} = -C_{ij}$, $f_i = -u_i$, and $g_j = -v_j$. Then (18) becomes the Kantorovich minimization (4), while $u_i + v_j \geq S_{ij}$ becomes the familiar dual inequality $f_i + g_j \leq C_{ij}$ from (6). Equality on matched pairs becomes (7).

Remark 6.3 (Why transfers change the mathematics). In ordinary stable marriage, the two comparisons in (8) are separate ordinal statements. In the assignment game, a transfer lets the pair pool its surplus, replacing two preference comparisons by the scalar inequality $u_i + v_j \geq S_{ij}$. That inequality is a Kantorovich dual constraint. This aggregation is the exact reason optimal transport enters the transferable-utility model but not the general Gale–Shapley model.

The continuous counterpart is equally important in economics. Quasilinear hedonic equilibria and stable matching with transferable utility can be cast as Monge–Kantorovich problems; see [Chiappori et al. \(2010\)](#) for existence, purity, and uniqueness questions in continuous type spaces.

7 Three integrality and duality mechanisms

The comparison can be summarized without collapsing the models into one another.

Model	Data and criterion	Convex geometry	Main algorithmic principle
Optimal assignment	Cardinal costs C_{ij} ; minimize total cost	$\mathcal{B}_n = \text{conv}\{P_\sigma\}$ by Birkhoff–von Neumann	Hungarian or auction primal–dual optimization
Stable marriage	Two ordinal preference profiles; exclude blocking pairs	$\mathcal{S} = \text{conv}\{P_\sigma : \sigma \text{ stable}\}$ by Vande Vate–Rothblum	Deferred acceptance; irreversible rejections and one-sided lattice optimum
Assignment game	Cardinal surplus S_{ij} with transfers; exclude profitable pair deviations	Primal Birkhoff polytope plus dual payoff inequalities $u_i + v_j \geq S_{ij}$	Linear-programming duality, complementary slackness, and market-clearing prices

The words *primal* and *dual* play different roles here. In the stable-marriage polytope, no-blocking is imposed by additional primal inequalities on P . In the assignment game, no-blocking is exactly dual feasibility for the payoffs. The latter is why Hungarian potentials can be read as equilibrium utilities or prices. Gale–Shapley’s tentative partners and rejections are not Kantorovich potentials, even though both algorithms maintain certificates that monotonically restrict the remaining choices.

8 Extensions and limits of the analogy

Unequal populations and outside options. For optimal transport, unequal cardinalities pose no conceptual difficulty: one uses rectangular coupling matrices with prescribed marginals. For stable marriage, one allows unmatched agents and declares being unmatched an outside option, or adds dummy agents. In the assignment game, dummy partners with zero surplus produce the same construction and make individual rationality explicit.

Many-to-one matching and capacities. College admissions replaces one-to-one matching by capacities on the receiving side. Deferred acceptance still applies under responsive

preferences. Its feasible incidence matrices resemble a capacitated transportation polytope, but stability again adds preference-dependent no-blocking constraints. The transportation constraints and the stability constraints should not be identified.

Fractional masses. Kantorovich transport naturally splits mass and accommodates arbitrary histograms. Stable marriage is instead an indivisible-agent model. Theorem 4.2 permits a fractional point only as a lottery over integral stable marriages; its integrality is special to bipartite stable matching with the stated preference assumptions. With ties, couples, nonbipartite roommates, or more general complementarities, existence and integrality can fail or require different stability notions.

Randomization. A bistochastic matrix can be interpreted as marginals of a lottery over matchings. Birkhoff–von Neumann says every such fractional assignment can be implemented by randomizing over deterministic assignments. If the matrix also lies in $\mathcal{S}$, Vande Vate–Rothblum says one can randomize using only stable assignments. This is stronger than checking expected preference scores: ex post every realized matching can be stable.

Continuous matching. In continuous populations, optimal transport retains its measure-theoretic meaning. Stable matching without transfers needs a suitable continuum market model and is not simply a transport problem. Under transferable utility and quasilinear preferences, however, Kantorovich dual potentials become indirect utilities or prices, and complementary slackness identifies equilibrium matches. This is the robust path from finite assignment games to continuous optimal transport.

9 Takeaway

The clean connection consists of two parallels and one exact equivalence.

1. **Parallel convex hulls.** Birkhoff–von Neumann convexifies all perfect matchings; Vande Vate–Rothblum convexifies the stable ones after adding linear no-blocking inequalities.
2. **Different objectives.** Ordinary stable marriage is an ordinal equilibrium problem, whereas Kantorovich assignment is a cardinal welfare problem. Neither generally implies the other.
3. **Exact transferable-utility bridge.** In the assignment game, core stability is Kantorovich dual feasibility, matched pairs are contact points of the dual constraint, and complementary slackness plus Birkhoff integrality produces an integral stable allocation.

Thus Gale–Shapley and optimal transport belong to the same broad geometry of bipartite couplings, but the assignment game is the precise point where their notions of stability and optimality coincide.

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